

Strain as Curvature:

A Geometric and Topological Framework for the Three-Dimensional Navier–Stokes Equations

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Abstract

We identify a distinguished geometric sector of the three-dimensional incompressible Navier–Stokes equations by decomposing the velocity gradient under $SO(3)$. The transverse-traceless strain S_{TT} is the unique multiplicity-one helicity- ± 2 sector, and it admits a natural curvature interpretation: for the effective metric $g = \delta + 2S$, it coincides with the linearized Einstein tensor. The Cauchy–Green tensor shows further that material deformation is governed by strain alone and is geodesic motion on $SL(3, \mathbb{R})/SO(3)$.

We prove that S_{TT} is protected against mixing by every bounded invertible rotation-covariant linear operator. Globally, the protected sector forms a non-orientable real rank-two bundle over the regular strain space, with quaternionic eigenframe monodromy $\pi_1(\mathcal{R}) \cong Q_8$ and nonzero Stiefel–Whitney class $w_1(E_{\mathbb{R}})$. The associated connection becomes singular at the discriminant locus, where strain eigenvalues coincide.

Finally, we reinterpret the Beale–Kato–Majda and Constantin–Fefferman regularity criteria through this geometry and formulate an eigenframe coherence conjecture linking their loss of coherence to possible vorticity concentration. No regularity theorem is claimed. What we offer instead is a single framework in which the curvature, the representation-theoretic protection, the global topology, and the singular degeneration of S_{TT} can all be studied together.

1 Introduction

The central difficulty of the three-dimensional incompressible Navier–Stokes equations is that no presently known estimate rules out indefinite concentration of the nonlinear dynamics. The usual approach is analytic, seeking control of critical norms. We take a complementary geometric route instead, and ask a more structural question: which part of the velocity gradient carries the intrinsic geometry of the deformation, and can that part be isolated once and for all?

The starting point is the classical geometric picture of fluid motion, from Arnold’s identification of ideal flow as geodesic motion on the diffeomorphism group [1] to the invariant analysis of the velocity gradient tensor [2, 3]. Within this picture we isolate a single distinguished sector, the transverse-traceless strain S_{TT} : the helicity- ± 2 component of the symmetric traceless strain. Four properties bind together around it. It reconstructs the velocity through the strain alone. It is a linearized curvature. It is protected against every covariant linear attempt to remove it. And, globally, it carries genuine topology.

Write

$$A = \nabla \mathbf{u} \in \mathfrak{sl}(3, \mathbb{R}), \quad A = \Omega + S, \quad \mathfrak{sl}(3, \mathbb{R}) = \underbrace{\mathfrak{so}(3)}_3 \oplus \underbrace{\mathfrak{p}}_5, \quad (1)$$

the Cartan decomposition, where Ω is the antisymmetric vorticity and S the symmetric traceless strain. Relative to a Fourier direction, S further decomposes into helicity sectors, with S_{TT} carrying helicity ± 2 .

The paper follows one continuous chain of reasoning,

$$\begin{array}{l} \text{strain} \longrightarrow \text{curvature} \longrightarrow S_{\text{TT}} \\ \longrightarrow \text{global topology} \longrightarrow \text{regularity} \end{array}$$

and each arrow is made precise in turn. Section 2 shows that the strain reconstructs the velocity and that the Navier–

Stokes nonlinearity closes within a finite spin algebra. Section 3 identifies S_{TT} with the linearized Einstein tensor of $g = \delta + 2S$. Section 4 proves that no covariant linear operator can remove it. Section 5 shows that, globally, the protected sector assembles into a real rank-two bundle with quaternionic monodromy and nonzero Stiefel–Whitney class. Section 6 translates the classical regularity criteria into this language and states an eigenframe coherence conjecture. No regularity theorem is proved anywhere in what follows; the aim is to make every other link in the chain precise, and to leave the coherence and degeneration of the strain eigenframe as the one remaining open question.

2 The strain as fundamental field

Work on $M = \mathbb{T}^3$, or on $\mathbb{R}^3$ with Schwartz-class decay, where the equations read $\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}$, $\nabla \cdot \mathbf{u} = 0$. Write Π_0, Π_1, Π_2 for the projections of $A = \nabla \mathbf{u}$ onto the trace, antisymmetric, and symmetric-traceless parts, so that $A = \Pi_0 A + \Pi_1 A + \Pi_2 A$ with $\Pi_0 A = 0$ by incompressibility, and the strain is $S = \Pi_2 A = \text{sym } \nabla \mathbf{u}$.

Theorem A (Reconstruction). *Let $\mathbf{u}$ be mean-zero and divergence-free on $\mathbb{T}^3$, with strain $S = \text{sym } \nabla \mathbf{u}$. Then*

$$\mathbf{u} = -2(-\Delta)^{-1} \nabla \cdot S, \quad (2)$$

and both the vorticity ω and the full gradient A are determined by S up to a spatially affine field.

Proof. Incompressibility gives $\nabla \cdot S = \frac{1}{2} \Delta \mathbf{u} + \frac{1}{2} \nabla (\nabla \cdot \mathbf{u}) = \frac{1}{2} \Delta \mathbf{u}$, which is (2). Equivalently, in Fourier space $\hat{A} = i \hat{\mathbf{u}} \otimes \mathbf{k}$, so $\hat{S} \mathbf{k} = \frac{i}{2} |\mathbf{k}|^2 \hat{\mathbf{u}}$, and inverting gives the same identity. The $\mathbf{k} = 0$ mode carries the affine ambiguity. $\square$

The velocity is therefore obtained from the strain by an operator of degree -1 : the strain is a complete variable for

the dynamics, at the price of a compatibility constraint taken up in Section 3.

For $X, Y \in \mathfrak{gl}(3) := \mathbb{R}^{3 \times 3}$, define $\langle X, Y \rangle_s := \Pi_2[s](XY)$ for $s \in \{0, 1, 2\}$, where $\Pi_2[s]$ projects onto the spin- s irreducible component of $\mathfrak{gl}(3) \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$. Matrix multiplication thus induces $SO(3)$ -equivariant bilinear couplings between the scalar, rotational, and symmetric-traceless sectors.

Theorem B (Spin- ≤ 2 closure). *Every term in the incompressible Navier–Stokes nonlinearity is a coupling $\langle \cdot, \cdot \rangle_s$ with $s \leq 2$.*

Proof. Under the adjoint action $X \mapsto RXR^{-1}$, $\mathfrak{gl}(3) \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$, of maximal spin 2, and matrix multiplication $m(X \otimes Y) = XY$ is $SO(3)$ -equivariant into $\mathfrak{gl}(3)$. Any spin- $s > 2$ summand of the domain therefore lies in $\ker m$, since the target has no irreducible component to receive it. Restricting to $A \in \mathbf{3} \oplus \mathbf{5}$, the surviving channels are exactly $s \in \{0, 1, 2\}$. $\square$

Projecting the differentiated equation onto these sectors recovers the pressure Poisson equation at $s = 0$, the vorticity equation

$$D_t \omega = \langle S, \Omega \rangle_1 + \nu \Delta \omega$$

at $s = 1$, and, at $s = 2$, the closed strain master equation

$$D_t S = -\langle S, S \rangle_2 - \frac{1}{4} \langle \Omega, \Omega \rangle_2 - \text{Hess}^{\text{dev}} p + \nu \Delta S, \quad (3)$$

where

$$\text{Hess}^{\text{dev}} p := \nabla^2 p - \frac{1}{3} (\Delta p) I$$

is the trace-free, anisotropic part of the pressure Hessian. In terms of the pressure Poisson source

$$\Delta p = \frac{1}{2} |\omega|^2 - \|S\|^2,$$

this nonlocal contribution can be written as

$$\text{Hess}^{\text{dev}} p = \mathcal{R}_2 \left(\frac{1}{2} |\omega|^2 - \|S\|^2 \right), \quad \mathcal{R}_2 = R_i R_j - \frac{1}{3} \delta_{ij},$$

with $R_i = \partial_i (-\Delta)^{-1/2}$. Thus the pressure enters the strain dynamics through a trace-free Riesz transform of the quadratic strain–vorticity source. Finally, using Theorem A, the vorticity is determined by S , so the equation closes on the strain field itself.

Proposition 1 (The unique nonlocal channel). *Aside from the viscous term $\nu \Delta S$, the deviatoric pressure Hessian is the only term in (3) coupling distinct spatial points: the unique nonlocal $\mathbf{1} \rightarrow \mathbf{5}$ channel.*

Proof. $\langle S, S \rangle_2$ and $\langle \Omega, \Omega \rangle_2$ are pointwise products. The multiplier symbol $\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij}$ of $\mathcal{R}_2$ is homogeneous of degree zero, hence genuinely nonlocal, and carries the spin-0 scalar $\frac{1}{2} |\omega|^2 - \|S\|^2$ into $\mathbf{5}$ through the traceless projection Π_2 . $\square$

This is the key structural fact. Drop the pressure Hessian and one is left with the restricted-Euler system of Vieillefosse and Cantwell [2, 3], which blows up in finite time for generic data. Whatever prevents singularity in the true equations, if anything does, must enter through this one nonlocal channel.

3 Geometry: strain as curvature

Let F solve $\dot{F} = AF$ with $F \in SL(3, \mathbb{R})$, and set $C = F^\top F$. Since C is invariant under $F \mapsto RF$ for $R \in SO(3)$, it removes the rotational part of the deformation and takes values in

$$P(3) = SL(3, \mathbb{R})/SO(3) = \{C = C^\top > 0, \det C = 1\},$$

a Hadamard manifold with isotropy $SO(3)$ at $C = I$ and tangent space $\mathfrak{p}$ [4]. The Cartan decomposition (1) is thus the infinitesimal form of the deformation geometry [1]: $\mathfrak{so}(3)$ generates rotations, while $\mathfrak{p}$ parametrizes volume-preserving changes of shape.

Theorem C (Geodesic deformation). *The Cauchy–Green tensor obeys $\dot{C} = 2F^\top S F$: deformation depends on the strain alone, with vorticity cancelling identically. The same strain governs vortex stretching,*

$$\frac{d}{dt} \log |\omega| = \langle \xi, S \xi \rangle = q_S(\xi), \quad \xi = \omega/|\omega|. \quad (4)$$

If S is constant in the co-moving frame, $C(t) = e^{2tS}$ is a geodesic on $P(3)$.

Proof. $\dot{C} = \dot{F}^\top F + F^\top \dot{F} = F^\top (A^\top + A) F = 2F^\top S F$, and dividing $\frac{d}{dt} |\omega|^2 = 2\langle \omega, S \omega \rangle$ by $|\omega|^2$ gives (4). When S is constant, $F = e^{tS}$ is a one-parameter subgroup generated by $S \in \mathfrak{p}$, hence a geodesic through the origin of $P(3)$ [4]. $\square$

Strain is thus the infinitesimal generator of shape deformation. The natural next question is which symmetric fields S can arise this way from an actual velocity field. Linearizing at $C = I$ gives the Saint-Venant compatibility condition: a traceless symmetric S equals $\text{sym} \nabla \mathbf{u}$ for some incompressible $\mathbf{u}$ exactly when the incompatibility $\text{inc}(S) := \nabla \times S \times \nabla$ vanishes, the linearized Bianchi identity for the metric $g = I + 2S$ [5].

Lemma 1 (S_{TT} as linearized curvature). *For $g_{ij} = \delta_{ij} + 2S_{ij}$, the linearized Einstein tensor equals the Saint-Venant incompatibility,*

$$G_{ij}^{(1)}[g] = \text{inc}(S)_{ij} = \varepsilon_{ikl} \varepsilon_{jmn} \partial_k \partial_m S_{ln},$$

and in Fourier space

$$\widehat{\text{inc}(S)}_{ij}(\mathbf{k}) = |\mathbf{k}|^2 \hat{S}_{ij}^{\text{TT}}(\hat{\mathbf{k}}). \quad (5)$$

Consequently: (i) $\text{inc}(S) = 0$ iff $S = \text{sym} \nabla \mathbf{u}$ for some divergence-free $\mathbf{u}$; (ii) S_{TT} is the unique part of S invariant under $S \mapsto S + \text{sym} \nabla \zeta$; and (iii) inc is elliptic of order two and injective on S_{TT} .

Proof. Linearizing the Riemann tensor of $g = \delta + 2\epsilon S$ and contracting gives the stated $G_{ij}^{(1)}$ [6, 5]. In Fourier space the double curl kills the longitudinal and mixed parts of $\hat{S}$, since ε_{ikl} is antisymmetric against the symmetric pair $k_k k_l$; on the transverse-traceless part, the only $SO(3)$ -equivariant degree-two scalar available is a multiple of $|\mathbf{k}|^2$, giving (5). Claims (i) through (iii) follow, using $\text{inc}(\text{sym} \nabla \zeta) = 0$ for any smooth ζ and $|\mathbf{k}|^2 \neq 0$ away from the origin. $\square$

In this sense S plays the role of a gauge potential, and S_{TT} is its curvature-bearing, gauge-invariant content. A gradient flow $\mathbf{u} = \nabla f$ has $S = \nabla^2 f$ and hence $S_{\text{TT}} \equiv 0$: it may look arbitrarily complicated in coordinates, but it carries no intrinsic linearized curvature at all.

4 The protection theorem

Having identified S_{TT} as the curvature-bearing sector, we ask whether it can be removed by a change of variables. Fix $\mathbf{k} \neq 0$; the relevant symmetry is the stabilizer $\text{Stab}(\mathbf{k}) = SO(2)_{\mathbf{k}} \subset SO(3)$, under which the fiber $\mathbf{3} \oplus \mathbf{5}$ decomposes into helicity sectors.

Lemma 2 (Helicity multiplicities). *Under $SO(2)_{\mathbf{k}}$, the fiber $\mathbf{3} \oplus \mathbf{5}$ has weight multiplicities $n_0 = 2$, $n_{\pm 1} = 2$, $n_{\pm 2} = 1$. The unique $m = \pm 2$ sector lies entirely in $\mathbf{5}$, and is exactly S_{TT} .*

Proof. The vector representation $\mathbf{3}$ decomposes into one $m = 0$ mode and one $m = \pm 1$ pair; the symmetric-traceless representation $\mathbf{5}$ into $m = 0$, $m = \pm 1$, and $m = \pm 2$. Summing gives the stated multiplicities, with $m = \pm 2$ the fully transverse part of the spin-2 sector. $\square$

The multiplicity-one nature of $m = \pm 2$ is the whole theorem in embryo: there is no equivalent sector for S_{TT} to mix into.

Theorem D (Protection). *Let $\mathcal{T}$ be a bounded, invertible, translation-invariant, $SO(3)$ -covariant linear operator on L^2 tensor fields (in particular any $SO(3)$ -covariant order-zero pseudodifferential operator). Then $\mathcal{T}$ preserves every helicity sector; acts on $m = \pm 2$ as multiplication by a nonzero scalar $c_{\pm 2}(|\mathbf{k}|)$, and can neither annihilate nor displace S_{TT} .*

Proof. Translation invariance gives a Fourier multiplier $\hat{\mathcal{T}}(\mathbf{k}) \in L(V_{\mathbb{C}}, V_{\mathbb{C}})$ on $V_{\mathbb{C}} = \mathbf{3}_{\mathbb{C}} \oplus \mathbf{5}_{\mathbb{C}}$. Covariance under $SO(3)$ gives $\rho(g)\hat{\mathcal{T}}(\mathbf{k})\rho(g)^{-1} = \hat{\mathcal{T}}(g\mathbf{k})$; taking $g \in \text{Stab}(\mathbf{k})$ shows $\hat{\mathcal{T}}(\mathbf{k})$ commutes with $\rho(SO(2)_{\mathbf{k}})$, and taking g to rotate among vectors of fixed length shows $\hat{\mathcal{T}}$ depends on $\mathbf{k}$ only through $|\mathbf{k}|$. Since the $m = \pm 2$ space is one-dimensional by Lemma 2, Schur's lemma gives $\hat{\mathcal{T}}(\mathbf{k})|_{m=\pm 2} = c_{\pm 2}(|\mathbf{k}|) \cdot \text{id}$. Were $c_{\pm 2}$ to vanish on a set of positive measure, invertibility would fail on fields supported there, a contradiction; so $c_{\pm 2} \neq 0$ almost everywhere, and the curvature sector survives every such transformation. $\square$

Corollary 1. *The Fourier transform, the Leray projector, and the Biot–Savart law are all bounded invertible $SO(3)$ -covariant multipliers, so by Theorem D none of them can touch S_{TT} . The familiar nonlinear substitutions fail for a complementary reason. Irrotational reductions such as Cole–Hopf [7, 8] and Madelung [9] impose $\mathbf{u} = \nabla f$, so $S = \text{Hess } f$ and hence $S_{\text{TT}} \equiv 0$ by Lemma 1(ii): they restrict to configurations where the curvature sector is simply absent, rather than transforming it away. An abelian spinor lift, such as the Clebsch or Hopf construction [10, 11], produces only a $U(1)$ curvature, which lives entirely in spin 0 and cannot reach spin 2. Any transform with a chance of reaching the protected sector must therefore leave this covariant linear framework altogether, most naturally by depending on the solution through the strain eigenframe itself.*

5 Topology of the protected sector

Everything so far is local: a statement about symbols at each $\mathbf{k}$. We now ask what happens globally, as the strain state itself varies.

Let $V = \text{Sym}_0^2(\mathbb{R}^3) \cong \mathbb{R}^5$, and let

$$\mathcal{R} = \{S \in V \setminus \{0\} : \sigma_1 > \sigma_2 > \sigma_3\}$$

be the regular stratum of strain states with distinct eigenvalues. Its complement,

$$\begin{aligned} \mathcal{D} &= \{S \in V \setminus \{0\} : \text{disc}(S) = 0\}, \\ \text{disc}(S) &= \frac{1}{2}(\text{tr } S^2)^3 - 27(\det S)^2, \end{aligned}$$

is the discriminant locus, a codimension-two set where at least two eigenvalues coincide. At each point of $\mathcal{R}$ the ordered orthonormal eigenframe is unique only up to independently reversing each axis, subject to the determinant staying +1: the residual symmetry group is

$$V_4 = \{I, r_1, r_2, r_3\} \subset SO(3),$$

where

$$r_1 = \text{diag}(1, -1, -1), \quad r_2 = \text{diag}(-1, 1, -1), \quad r_3 = \text{diag}(-1, -1, 1),$$

the Klein four group.

Lemma 3 (Frame monodromy). *$\mathcal{R}$ is homotopy equivalent to $SO(3)/V_4$, and*

$$\pi_1(\mathcal{R}) \cong Q_8,$$

the quaternion group of order eight.

Proof. The eigenvalue map $S \mapsto (\sigma_1, \sigma_2, \sigma_3)$ presents $\mathcal{R}$ as a fiber bundle over the open wedge $\Sigma = \{\sigma_1 + \sigma_2 + \sigma_3 = 0, \sigma_1 > \sigma_2 > \sigma_3\}$ of the traceless plane, an open angular sector with its apex removed, hence contractible. To conclude that $\mathcal{R}$ is homotopy equivalent to a single fiber, we invoke Ehresmann's fibration theorem [12], which requires the eigenvalue map to be a proper submersion on $\mathcal{R}$. This holds because, for any compact $K \subset \Sigma$, the preimage in $\mathcal{R}$ has bounded Frobenius norm and stays a positive distance from $\mathcal{D}$, since K lies in the interior of the open wedge, away from its boundary where eigenvalues collide; a closed, bounded set at positive distance from the removed locus is compact. Ehresmann's theorem then gives a locally trivial fibration over Σ , and since Σ is contractible, $\mathcal{R}$ is homotopy equivalent to any one fiber.

Fix distinct $\sigma_1 > \sigma_2 > \sigma_3$ and write $D = \text{diag}(\sigma_1, \sigma_2, \sigma_3)$. The corresponding fiber consists of $S = RDR^T$, $R \in SO(3)$, and two rotations give the same S exactly when $R^{-1}R'$ centralizes D ; since the eigenvalues are distinct, this centralizer is exactly V_4 . So this fiber is diffeomorphic to $SO(3)/V_4$, confirming the identification above.

It remains to compute $\pi_1(SO(3)/V_4)$. Under the spin cover $p : SU(2) \rightarrow SO(3)$, with unit quaternions acting on $\text{Im } \mathbb{H}$ by conjugation and kernel $\{\pm 1\}$, a direct check with the quaternion relations shows $p^{-1}(V_4) = \{\pm 1, \pm i, \pm j, \pm k\}$. The same fact follows structurally, without computation: the preimage of the dihedral group of order $2n$ under the spin cover is always the binary dihedral group of the same order, presented as $\langle a, b \mid a^{2n} = 1, b^2 = a^n, bab^{-1} = a^{-1} \rangle$ [13]. Here V_4 is the dihedral group of order four ($n = 2$), giving the presentation $\langle a, b \mid a^4 = 1, b^2 = a^2, bab^{-1} = a^{-1} \rangle$, which is exactly the standard presentation of Q_8 under $a = i$, $b = j$. This rules out the abelian alternatives of order eight (such as $\mathbb{Z}_4 \times \mathbb{Z}_2$) that a bare cardinality count would leave open.

Since $\{\pm 1\} \trianglelefteq Q_8 \trianglelefteq SU(2)$ and $Q_8/\{\pm 1\} \cong V_4$, the spin cover descends to $SO(3)/V_4 \cong SU(2)/Q_8$. The group Q_8 acts on $SU(2) \cong S^3$ by left multiplication, freely, since only the identity quaternion fixes a point, and properly discontinuously, being a finite group acting by homeomorphisms on a Hausdorff space. So $SU(2) \rightarrow SU(2)/Q_8$ is the universal cover, and $\pi_1(SU(2)/Q_8) \cong Q_8$. Combining the three steps, $\pi_1(\mathcal{R}) \cong \pi_1(SO(3)/V_4) \cong \pi_1(SU(2)/Q_8) \cong Q_8$. $\square$

Remark 1. A small loop linking the discriminant face $\{\sigma_1 = \sigma_2 > \sigma_3\}$ induces, in $SO(3)$, a π rotation about $\mathbf{e}_3$; its lift is $k \in Q_8$. Linking $\{\sigma_1 > \sigma_2 = \sigma_3\}$ gives the lift $i \in Q_8$. Both have order four, since $k^2 = -1$. The physical eigenframe returns to itself after two traversals of either loop, a full 2π rotation, and it is tempting to read this as period two. But as a based loop in $\mathcal{R}$ it is null-homotopic only after four traversals. This is the Dirac belt trick, the same periodicity carried by a spin- $\frac{1}{2}$ particle, and it is the reason the true monodromy is the non-abelian Q_8 rather than $\mathbb{Z}_2$.

Given the eigenframe $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$, define

$$\mathbf{u} = \frac{1}{2}(\mathbf{e}_1 \otimes \mathbf{e}_1 - \mathbf{e}_2 \otimes \mathbf{e}_2), \quad \mathbf{v} = \frac{1}{2}(\mathbf{e}_1 \otimes \mathbf{e}_2 + \mathbf{e}_2 \otimes \mathbf{e}_1),$$

set $E_{\mathbb{R}}|_S = \text{span}_{\mathbb{R}}\{\mathbf{u}, \mathbf{v}\} \subset V$, and let $\Psi = \mathbf{u} + i\mathbf{v}$ be the associated weight- (-2) complex vector.

Proposition 2. $E_{\mathbb{R}} \rightarrow \mathcal{R}$ is a well-defined real rank-two bundle, but it does not admit a globally consistent complex line structure. The Q_8 generators act by $k : \Psi \mapsto \Psi$ and $i : \Psi \mapsto \bar{\Psi}$. So $\langle k \rangle \cong \mathbb{Z}/4$ preserves the complex line $\mathbb{C}\Psi$, while i exchanges it with its conjugate.

Proof. Under k , $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \mapsto (-\mathbf{e}_1, -\mathbf{e}_2, \mathbf{e}_3)$; since $\mathbf{u}, \mathbf{v}$ are each bilinear in $\mathbf{e}_1, \mathbf{e}_2$, both are unchanged and $\Psi \mapsto \Psi$. Under i , $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3) \mapsto (\mathbf{e}_1, -\mathbf{e}_2, -\mathbf{e}_3)$, so $\mathbf{u} \mapsto \mathbf{u}$ but $\mathbf{v} \mapsto -\mathbf{v}$, and $\Psi \mapsto \bar{\Psi}$. In either case $\text{span}_{\mathbb{R}}\{\mathbf{u}, \mathbf{v}\}$ is unchanged, so $E_{\mathbb{R}}$ is well defined even though $\mathbb{C}\Psi$ is not. $\square$

Theorem E (Global protection). $E_{\mathbb{R}} \rightarrow \mathcal{R}$ has structure group $O(2)$, and this cannot be reduced to $SO(2)$ over all of $\mathcal{R}$. Its Stiefel–Whitney class $w_1(E_{\mathbb{R}}) \in H^1(\mathcal{R}; \mathbb{Z}_2) = \text{Hom}(Q_8, \mathbb{Z}_2)$ is the orientation character $Q_8 \twoheadrightarrow Q_8/\langle k \rangle \cong \mathbb{Z}_2$, which is nonzero. So $E_{\mathbb{R}}$ is non-orientable and nontrivial as a bundle. No bounded invertible $SO(3)$ -covariant transform can trivialize it, since Theorem D shows any such transform acts fiberwise as a scalar depending only on $|\mathbf{k}|$, and a scalar cannot alter w_1 .

Proof. By Proposition 2, k preserves the orientation of the basis $(\mathbf{u}, \mathbf{v})$, while i sends $(\mathbf{u}, \mathbf{v}) \mapsto (\mathbf{u}, -\mathbf{v})$, which reverses it. The orientation character is therefore trivial on $\langle k \rangle$ and non-trivial on the complementary coset, giving $Q_8 \twoheadrightarrow Q_8/\langle k \rangle \cong \mathbb{Z}_2$; since this character is nonzero, $w_1(E_{\mathbb{R}}) \neq 0$, so the bundle is non-orientable and admits no reduction to $SO(2)$. $\square$

Remark 2. As S approaches $\mathcal{D}$, the eigenframe becomes ill defined and the connection $A = \mathbf{e}_1 \cdot d\mathbf{e}_2$ diverges as $|A| \sim \delta^{-1/2}$, where δ is the normalized discriminant, reflecting $|\hat{R}| \sim (\sigma_1 - \sigma_2)^{-1}$. The discriminant locus is exactly where the eigenframe, the connection, and $E_{\mathbb{R}}$ itself all degenerate together.

6 Consequences for regularity

Integrating (4) gives $|\omega(t)| = |\omega_0| \exp \int_0^t q_S(\xi) ds$, so the Beale–Kato–Majda criterion [14] requires this integral to diverge: the $m = 0$ contraction of S along ξ , the sector complementary to S_{TT} , must stay sustained in the extensional region. The Constantin–Fefferman criterion [15, 16] instead imposes coherence on the vorticity direction ξ ; in the language developed here, this constrains the alignment between ξ and the strain eigenframe, the very frame carrying S_{TT} . Together with the restricted-Euler blow-up isolated in Proposition 1, both criteria probe different projections of the same underlying curvature.

Conjecture 1 (Eigenframe coherence). Let $Q = B(\mathbf{x}_0, r) \times (t_0 - r^2, t_0)$ be a parabolic cylinder, with spatial radius r and the corresponding parabolic time scale r^2 . Say that the strain eigenframe is (C_0, r) -coherent on Q if the eigenvalues of S are simple throughout Q , so that the orthonormal eigenvectors $\mathbf{e}_i$ are well-defined, and

$$\|\mathbf{e}_i(\mathbf{x}, t) - \mathbf{e}_i(\mathbf{y}, s)\| \leq C_0(|\mathbf{x} - \mathbf{y}| + |t - s|^{1/2}), \quad i = 1, 2, 3,$$

for all $(\mathbf{x}, t), (\mathbf{y}, s) \in Q$. Thus the eigenframe varies in a uniformly Hölder-continuous manner with respect to the natural parabolic scaling of Navier–Stokes. If a fixed constant C_0 gives such coherence on every parabolic cylinder in a region where $\|\omega\|_{\infty} \geq \lambda \gg 1$, then

$$\int_0^T \|\omega\|_{\infty} ds < \infty,$$

and the solution is regular on $[0, T]$.

This strengthens Constantin–Fefferman, which controls only the cruder spin-1 direction ξ , by demanding coherence of the full gauge-invariant eigenframe. The hypothesis fails exactly where eigenvalues cease to be simple, on $\mathcal{D}$, the same locus where $\text{inc}(S)$ concentrates and the connection diverges: loss of eigenframe coherence and concentration of curvature are two faces of a single potential singular mechanism. Eigenframe alignment statistics are directly measurable in simulation [17, 18, 19], offering an empirical test of the conjecture well before any proof is in hand.

7 Conclusion

Three-dimensional incompressible Navier–Stokes is, on this reading, a spin-2 theory. The velocity gradient is a linearized metric, and its transverse-traceless strain is that metric’s linearized Einstein tensor (Lemma 1). Fluid deformation is geodesic motion on $SL(3, \mathbb{R})/SO(3)$ (Theorem C). The curvature sector is the unique multiplicity-one helicity eigenspace of every covariant linear operator (Theorem D), and globally it assembles into a real rank-two bundle obstructed by $w_1(E_{\mathbb{R}}) \neq 0$ under the quaternionic frame monodromy $\pi_1(\mathcal{R}) \cong Q_8$ (Section 5).

No regularity theorem has been proved. What has been pinned down, rigorously and globally, is the one geometric object any such proof will have to act on: the transverse-traceless strain, together with the exact place, the discriminant

locus, where that object, its eigenframe, and its connection all break down at once. The regularity conjecture above turns this into a single question: whether sufficient coherence of the strain eigenframe in regions of large vorticity is enough to prevent blow-up.

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